



Unit - 4

Networking for IoT



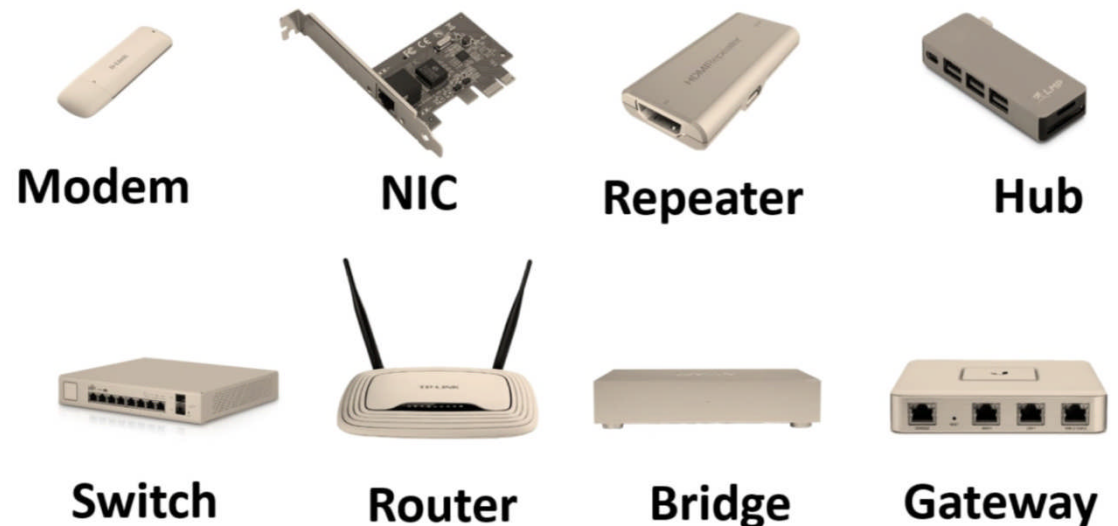


Objective

- Networking Devices
- Types of Networking Devices
- Local and Personal Area Network (LAN/PAN) for IoT
- IoT WAN
- IoT NODE
- IoT Gateway
- IPv4 and IPv6
- Multi Homing
- IoT Protocol

What are Networking Devices?

- Network devices, or networking hardware, are physical devices that are required for communication and interaction between hardware on a computer network.



Types of Network Devices

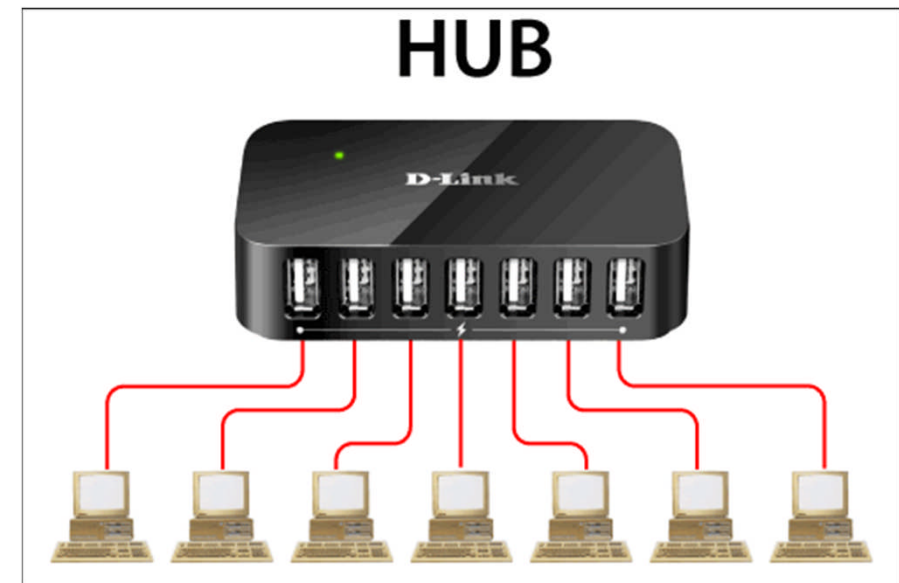
Types of Networking Devices

Here is the common network device list:

- Hub
- Switch
- Router
- Bridge
- Gateway
- Modem
- Repeater
- Access Point

Hub

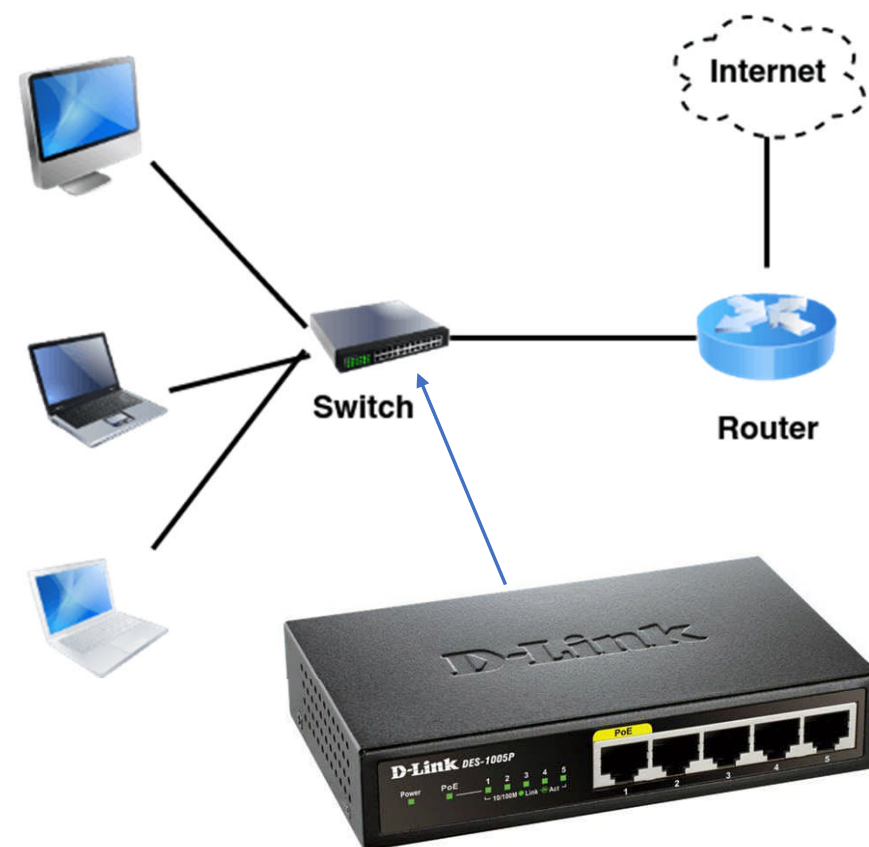
- A hub is basically a multiport repeater.
- A hub connects multiple wires coming from different branches, for example, the connector in star topology which connects different stations. Hubs cannot filter data, so data packets are sent to all connected devices.
- Hubs categories in three categories, Active hub, Passive Hub and Intelligent hub



<https://www.tutorialandexample.com/wp-content/uploads/2020/10/image-196.png>

Switch

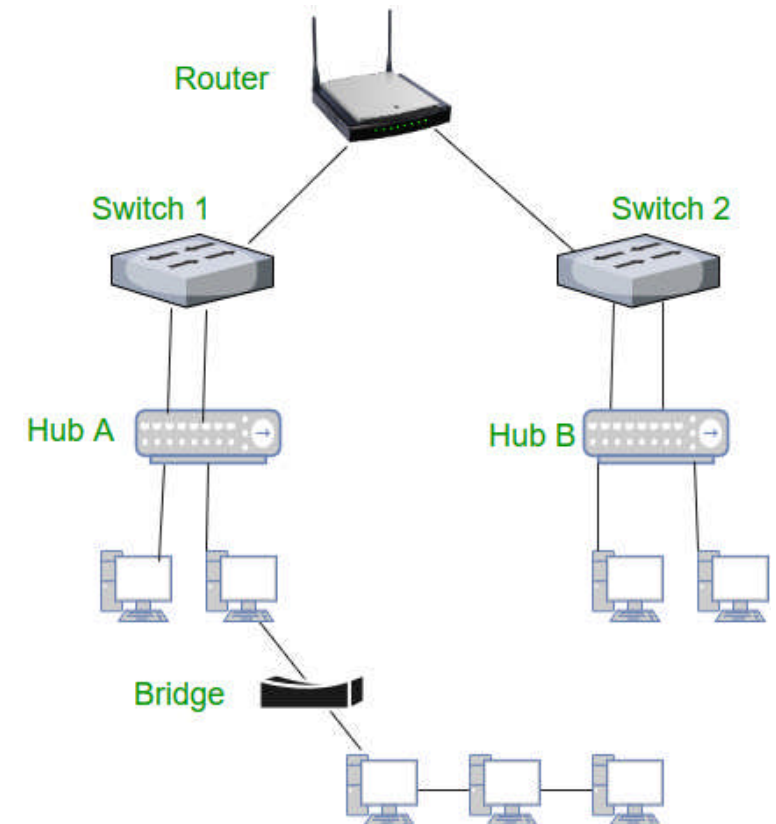
- A switch is a multiport device that improves network efficiency. The switch maintains limited routing information about nodes in the internal network, and it allows connections to systems like hubs or routers.
- switches can read the hardware addresses of incoming packets to transmit them to the appropriate destination.



https://www.nicepng.com/downloadpng/u2w7y3o0r5y3e6a9_computer-network-devices-switch-computer-network-device/

Router

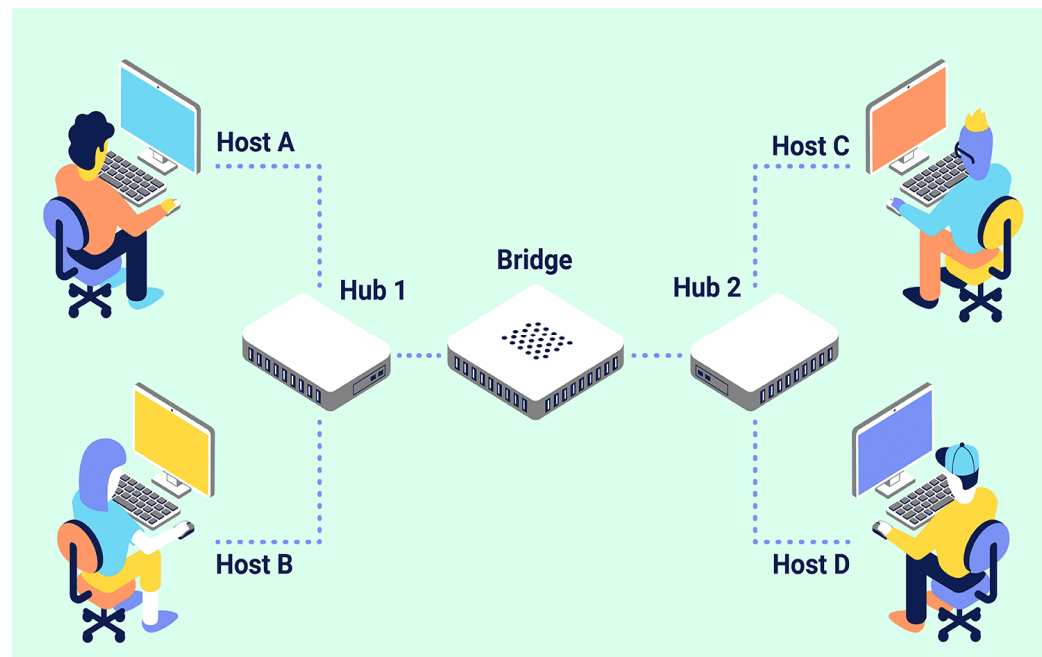
- A router is a device like a switch that routes data packets based on their IP addresses.
- The router is mainly a Network Layer device. Routers normally connect LANs and WANs together and have a dynamically updating routing table based on which they make decisions on routing the data packets.



: <https://www.geeksforgeeks.org/network-devices-hub-repeater-bridge-switch-router-gateways/>

Bridge

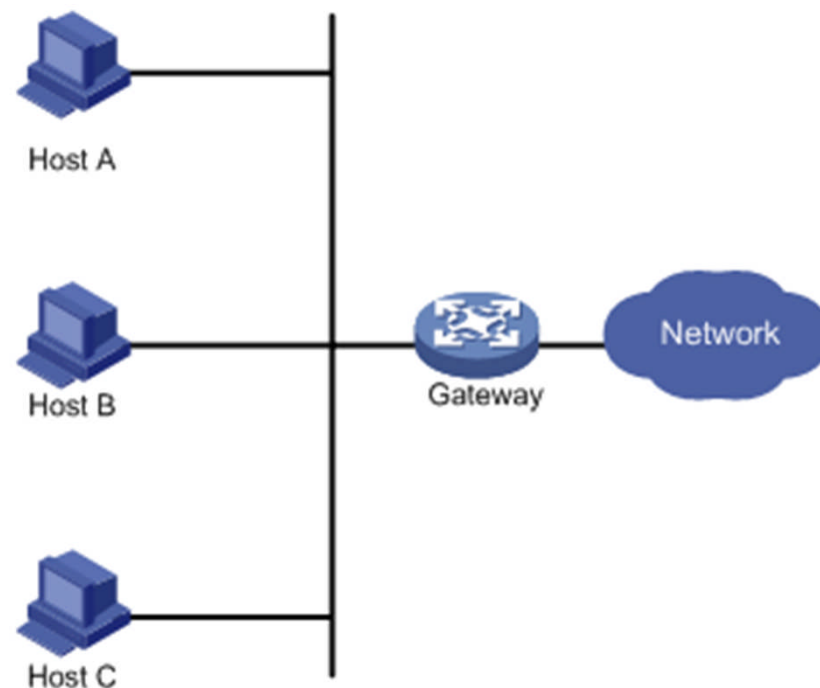
- A bridge operates at the data link layer.
- A bridge is a repeater, with add on the functionality of filtering content by reading the MAC addresses of source and destination.
- It is also used for interconnecting two LANs working on the same protocol.
- Types of Bridge
 - Transparent Bridge
 - Source routing Bridge



[https://www.lifewire.com/thmb/Xkm1rc4iiMVjeoWz5N5cX19d07w=/1500x1000/filters:fill\(auto,1\)/how-network-bridges-work-816357-eea65d0c474e429883dca2bfdcc83c30.png](https://www.lifewire.com/thmb/Xkm1rc4iiMVjeoWz5N5cX19d07w=/1500x1000/filters:fill(auto,1)/how-network-bridges-work-816357-eea65d0c474e429883dca2bfdcc83c30.png)

Gateway

- A gateway, as the name suggests, is a passage to connect two networks together that may work upon different networking models.
- They basically work as the messenger agents that take data from one system, interpret it, and transfer it to another system.
- Gateways are also called protocol converters and can operate at any network layer.



https://techhub.hpe.com/eginfolib/networking/docs/switches/5940/5200-1011b_hi-avail_cg/content/images/image32.png

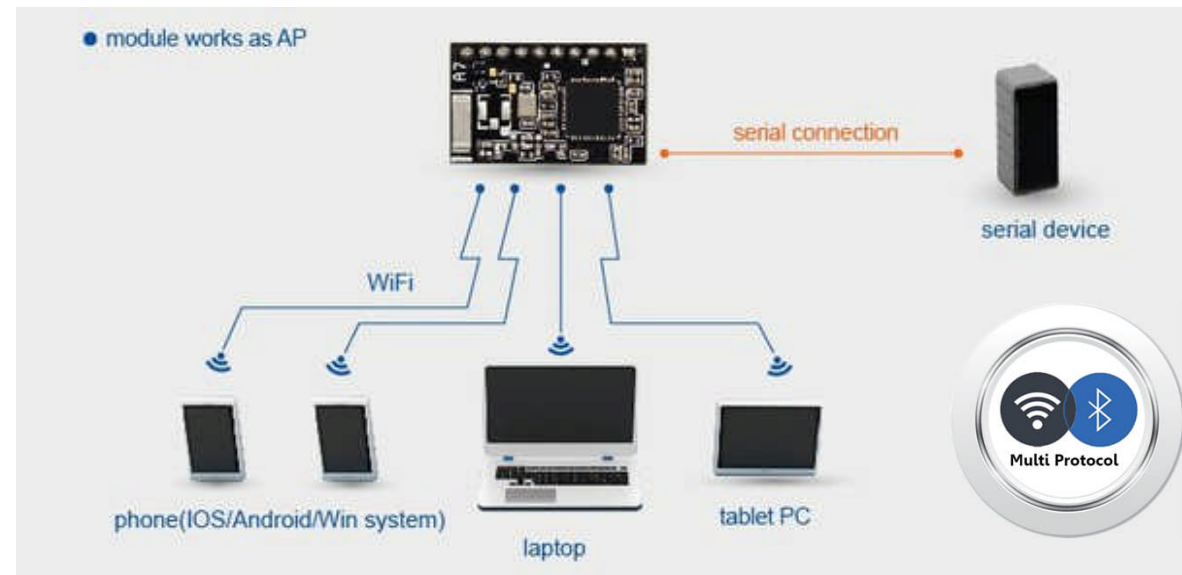
Repeater

- A repeater operates at the physical layer.
- Its job is to regenerate the signal over the same network before the signal becomes too weak or corrupted so as to extend the length to which the signal can be transmitted over the same network.



PAN/LAN

- Networks that cover fairly short distances are called personal area networks (PAN) and local area networks (LAN).
- PAN and LAN networks are considered to be fairly cost-effective, but the transfer of data can sometimes be unreliable.
Ex. Bluetooth, Wi-fi

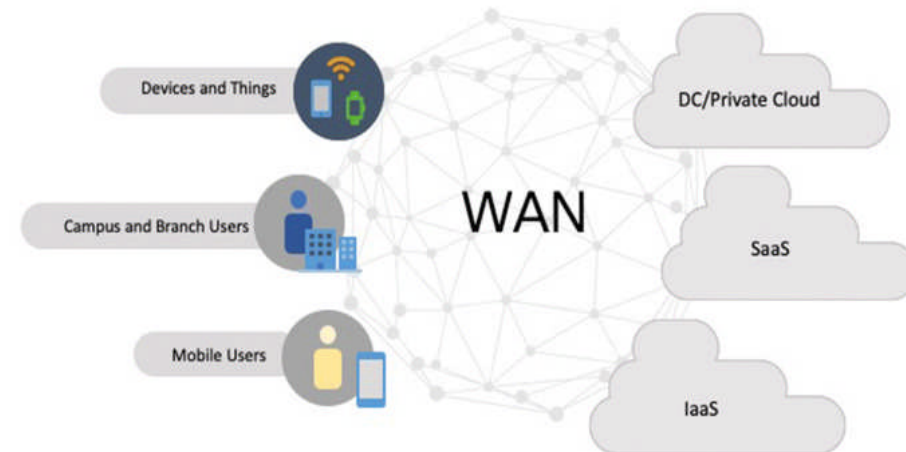


<https://www.quora.com/How-does-the-Internet-of-Things-work-in-a-LAN-network>

IoT WAN

- A wide area network (also known as WAN), is a large network of information that is not tied to a single location.
- WANs can facilitate communication, the sharing of information and much more between devices from around the world through a WAN provider.
- An Internet of Things (IoT) gateway is a device which serves as the connection point between IoT devices and the cloud.

Today, things have changed completely



<https://link.springer.com/article/10.1007/s40012-020-00277-5>

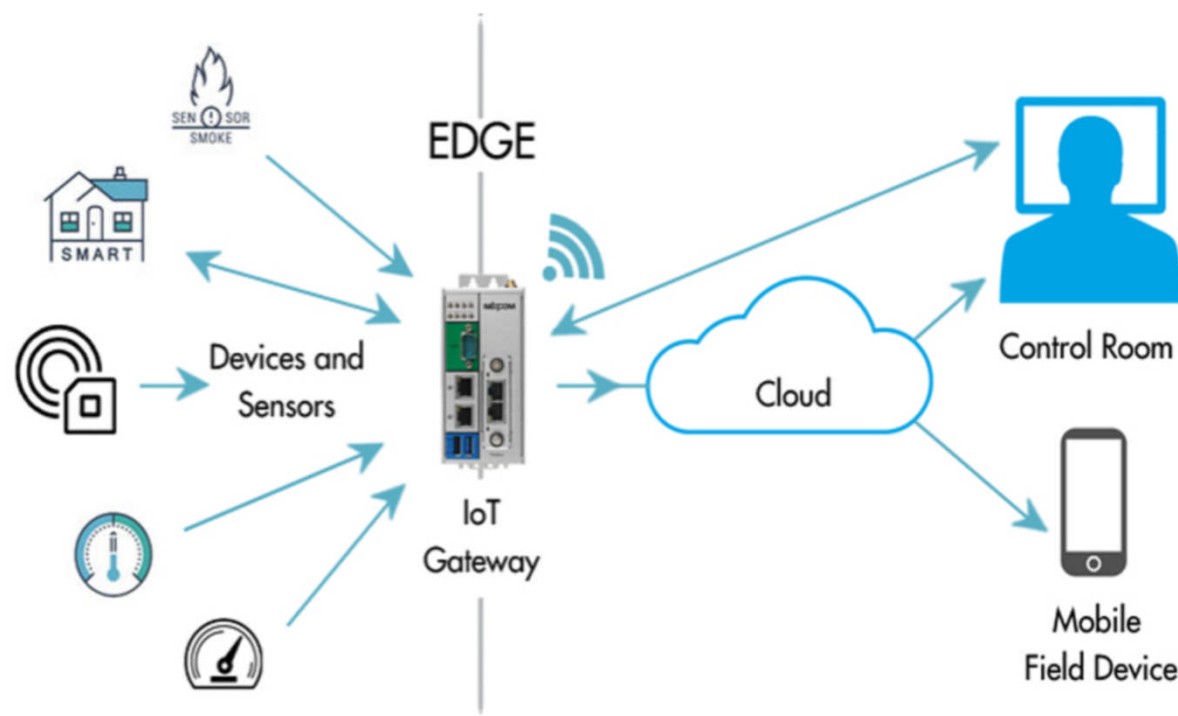
IoT Node

- The most numerous type of device in the IoT can be referred to as the node.
- These are all the exciting devices that are providing sensor data, or devices that are being controlled from the cloud.
- This means things like door locks, security sensors, temperature sensors, and more



IoT Gateway

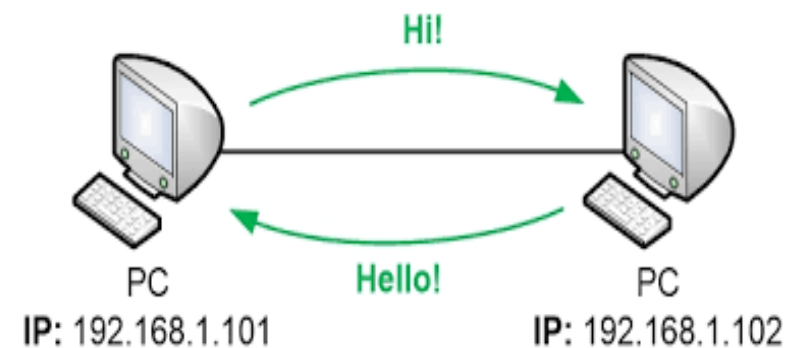
- Secure connection between Device and Cloud
- Data/information sharing between cloud and device
- Smart Gateways perform edge analytics on data produce by IoT device



https://www.ezenroute.com/assets/images/gif/iot_gateway_img_1.jpg

IPv4 and IPv6

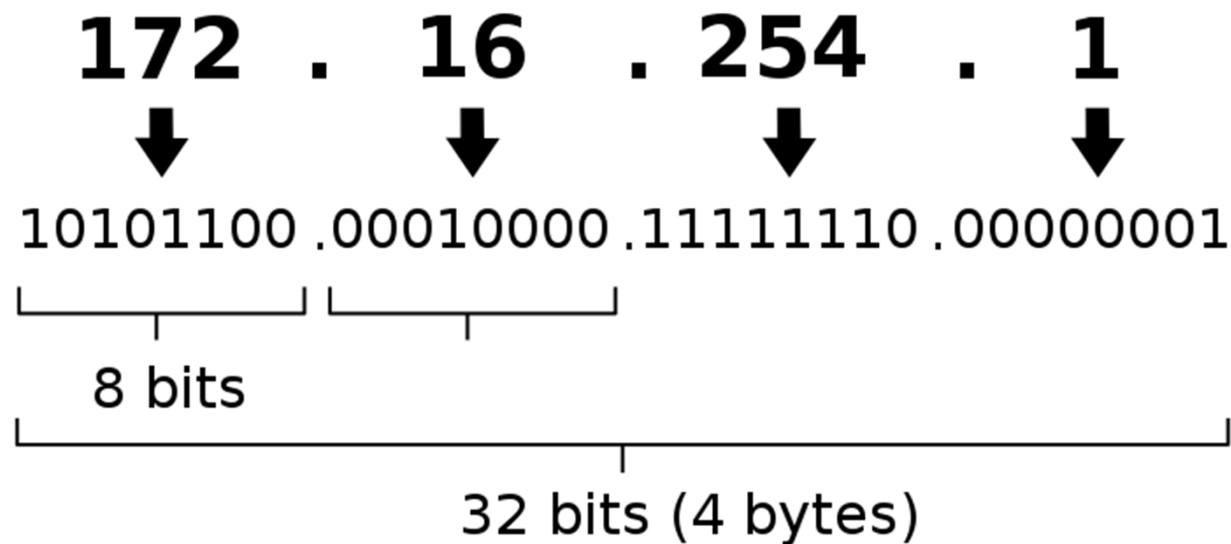
- **What is IP?**
- An IP (Internet Protocol) address is a numerical label assigned to each device connected to a computer network that uses the IP protocol for communication.
- An IP address acts as an identifier for a specific device on a particular network. The IP address is also called an IP number or Internet address.



<https://www.homenethowto.com/basics/ip-addresses/>

IPv4 – Internet Protocol Version 4

IPv4 address in dotted-decimal notation

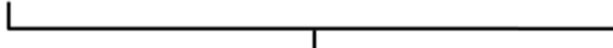


https://upload.wikimedia.org/wikipedia/commons/thumb/7/74/Ipv4_address.svg/750px-Ipv4_address.svg.png

IPv6 – Internet Protocol Version 6

An IPv6 address (in hexadecimal)

2001:0DB8:AC10:FE01:0000:0000:0000:0000

↓ ↓ ↓ ↓ 
2001:0DB8:AC10:FE01:: Zeroes can be omitted


 0010000000000001:0000110110111000:1010110000010000:111111000000001:
 0000000000000000:0000000000000000:0000000000000000:0000000000000000

https://commons.wikimedia.org/wiki/File:Ipv6_address.svg

IPv4

Deployed 1981

32-bit IP address

4.3 billion addresses

Addresses must be reused and masked

Numeric dot-decimal notation

192.168.5.18

DHCP or manual configuration

IPv6

Deployed 1998

128-bit IP address

7.9×10^{28} addresses

Every device can have a unique address

Alphanumeric hexadecimal notation

50b2:6400:0000:0000:6c3a:b17d:0000:10a9

(Simplified - 50b2:6400::6c3a:b17d:0:10a9)

Supports autoconfiguration

Multi Homing

- Multi-homing is a method of configuring one computer, called the host, with more than one network connection and IP address. The multi-homed method provides enhanced and reliable Internet connectivity without compromising efficient performance.
- *Why multi-homing in IOT?*
- In IOT particular Node or an IoT device or the sub network, IoT sub network can be connected with multiple networks for improving the reliability.
- So, basically multi-homing is a concept that is used for improving the overall liability of the network in that way.

IoT Protocol Stack

OSI Model	TCP/IP Model	Layer functions	Protocols
Application	Application	Directly interact with and support user's applications	MQTT, HTTPS, AMQP, CoAP
Presentation			
Session			
Transport	Transport	Handle reliability, flow control, congestion avoidance, and error correction	TCP, UDP
Network	Internet	Involve logical addressing and define how data is routed from sources to final destination hosts identified by IP addresses (traffic directing)	IP (e.g. IPv6, 6LoW-PAN...)
Data Link	Network access & physical	Define the physical connection of end devices to the network	LPWAN, WiFi, LTE, BLE, Zigbee
Physical			

Link Layer Protocol

802.3 Ethernet

802.11 Wi-Fi

802.16 WiMAX

802.15.4 LR-WPAN

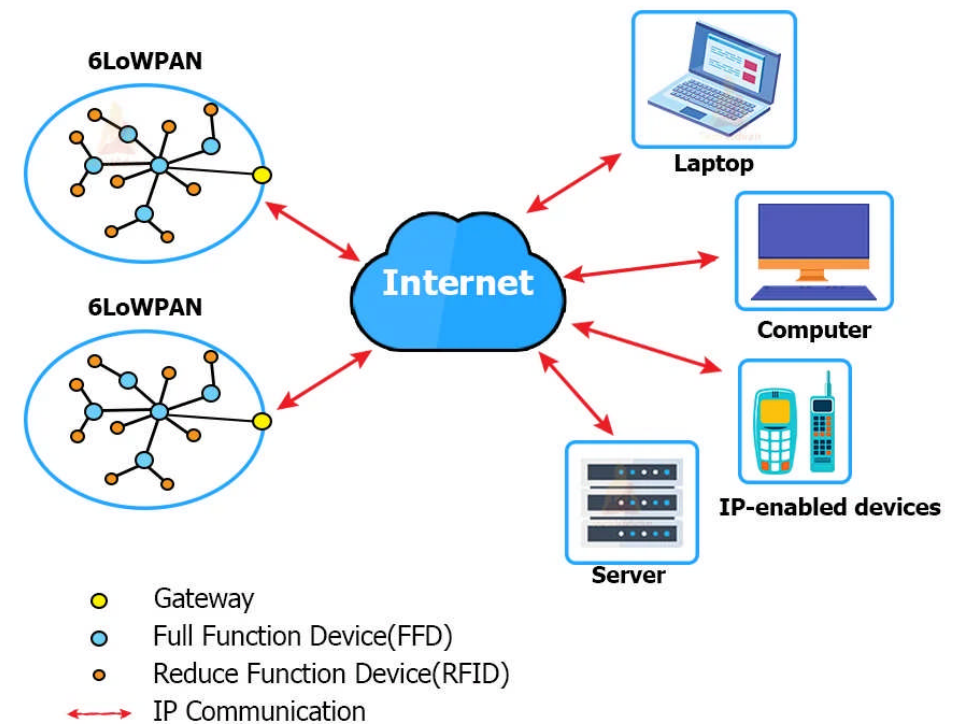
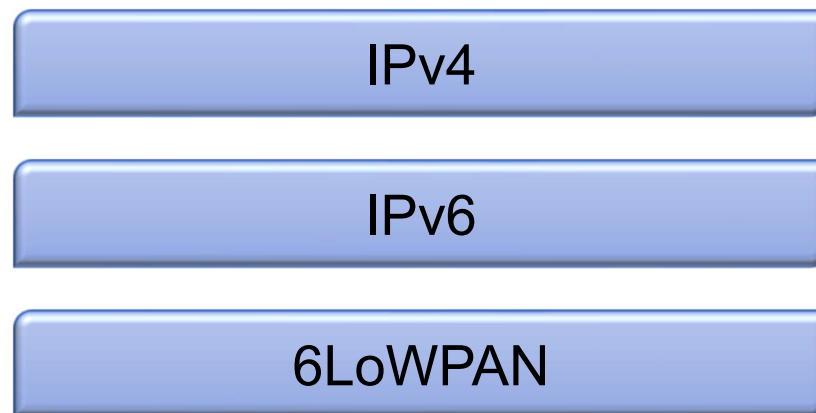
2G/3G/4G Mobile Comm.



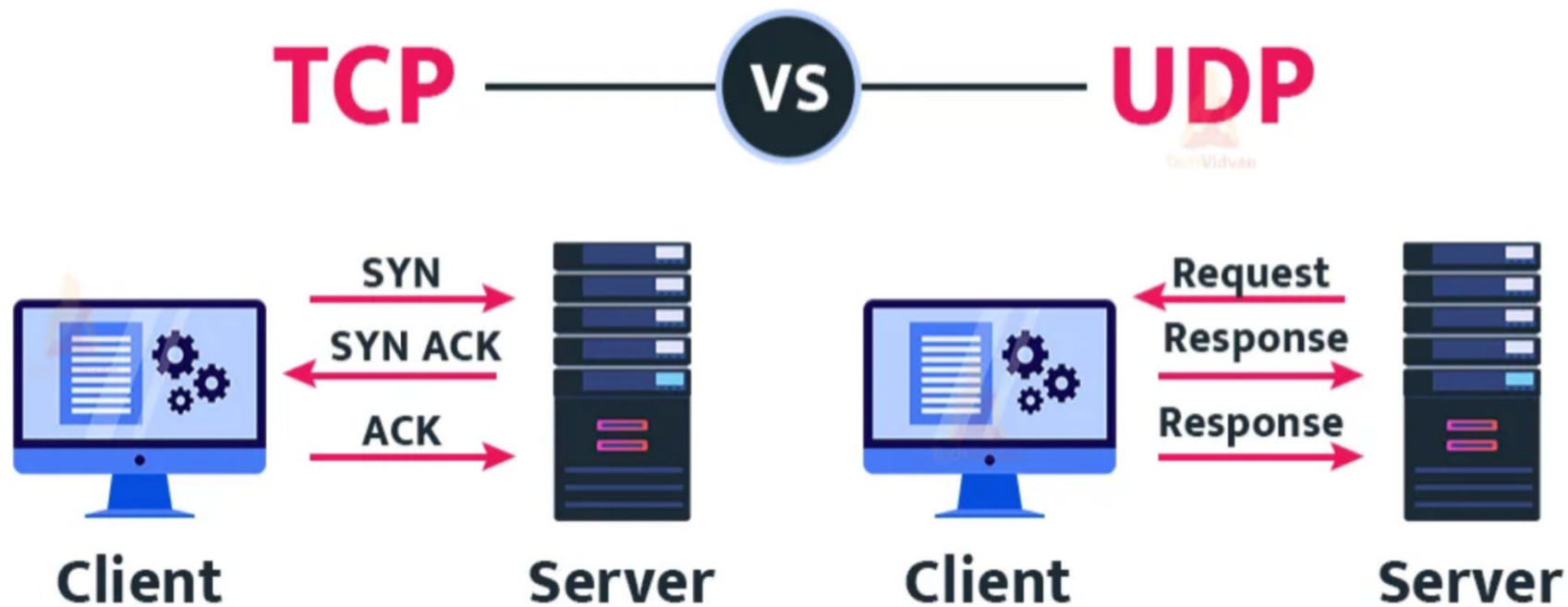
<https://techvidvan.com/tutorials/wp-content/uploads/sites/2/2021/10/Link-Layer-protocols.webp>

Network / Internet Layer Protocol

6LoWPAN Architecture



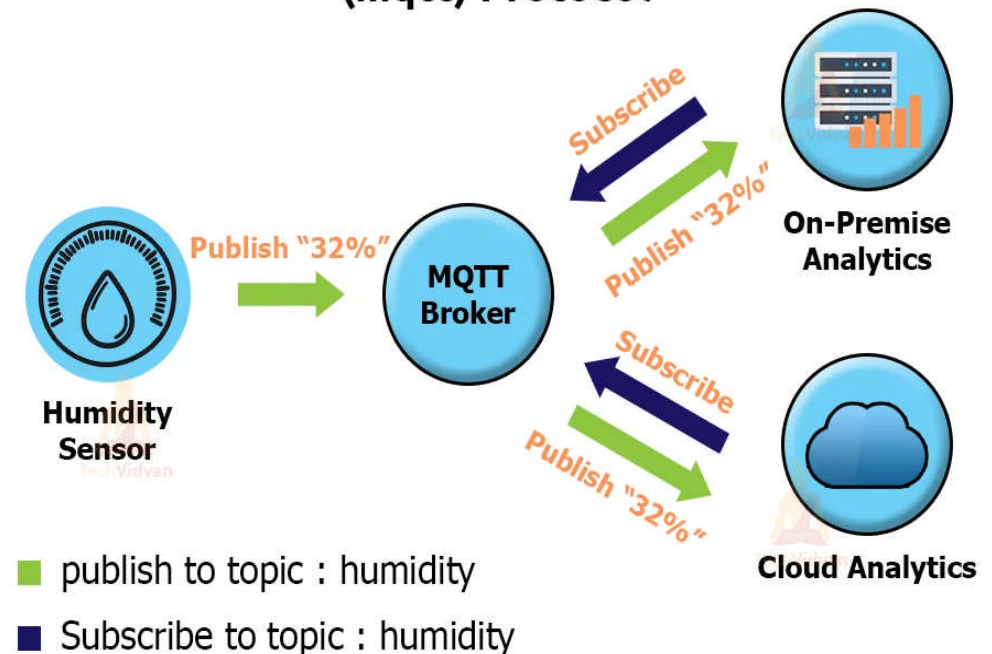
Transport Layer Protocol



Application Layer Protocol

- **MQTT**
- Message Queue Telemetry Transport it is a lightweight message protocol based on public -subscribe model.
- MQTT uses a client server Architecture by the clients such as an IoT device connect to the server also called the MQTT broker and publishers' message to topic on the server.

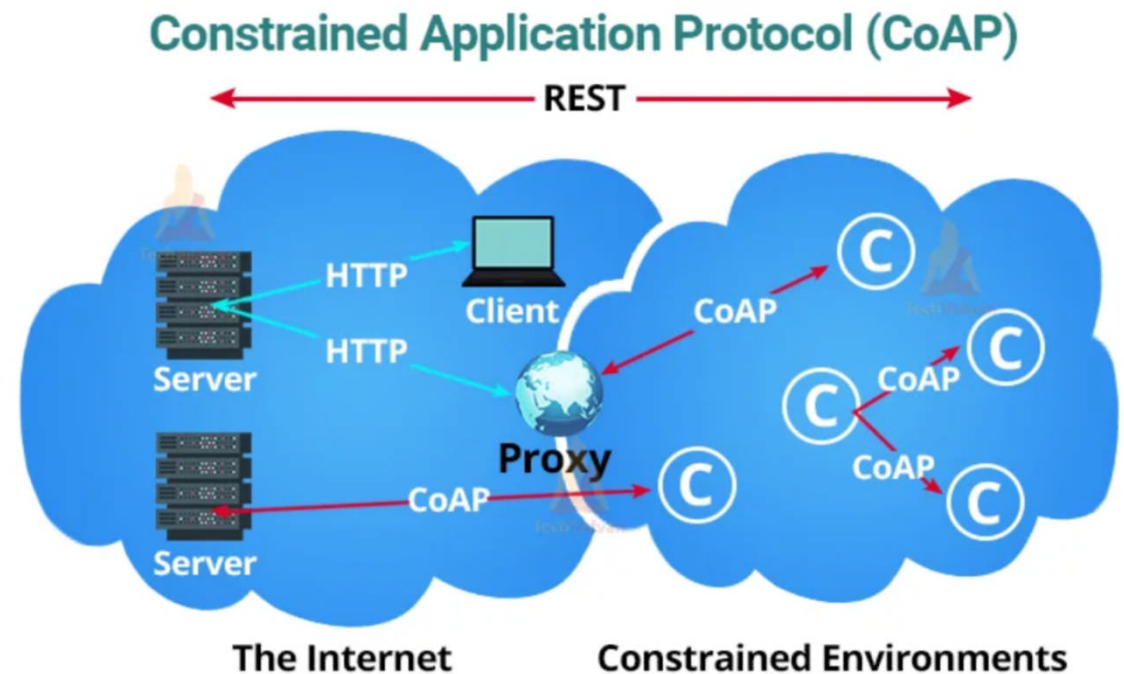
Message Queuing Telemetry Transport (mqtt) Protocol



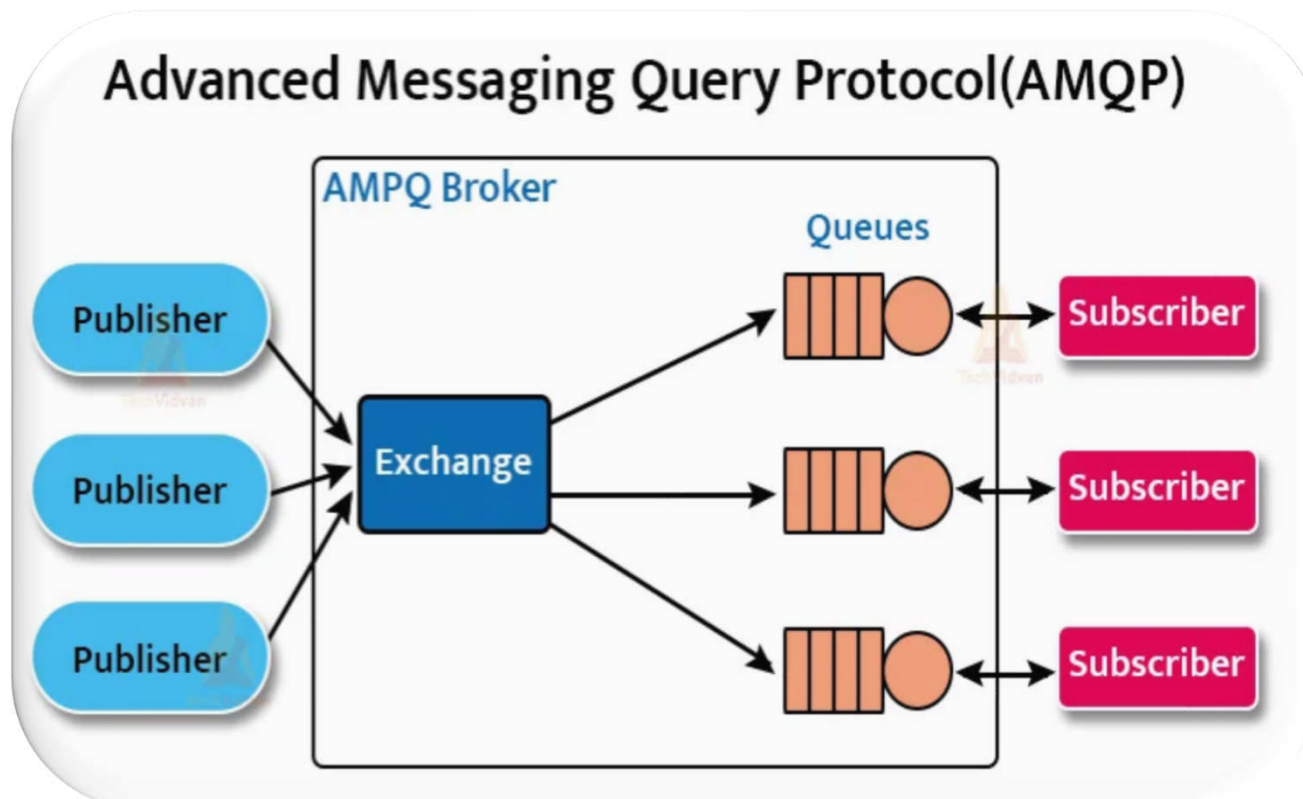
<https://techvidvan.com/tutorials/wp-content/uploads/sites/2/2021/10/message-queuing-telemetry-transportmqtt.webp>

Application Layer Protocol - CoAP

- In CoAP, a smart “thing” is capable of acting both as a **server** and **client**.
- A client sends a request for a server resource through a method code.
- The server replies with a respond code.



Application Layer Protocol - AMQP



- it is an open application layer protocol for business messaging.
- AMQP support point to point and publish - subscribe model routing and queuing



Reference

- <https://techvidvan.com/tutorials/communication-protocols-in-iot/>
- <https://www.arubanetworks.com/en-in/solutions/internet-of-things/>